

Ass_DFT_1.0

Refer to **Ass_Syn_2.0**, after synthesizing **your UART_TX**. You have to **insert DFT logic** and generate **Technology Dependent gate level netlists post-dft in Verilog format**

Steps: -

1. Perform RTL preparation for the **UART TX** (Top Module)
 - Add scan ports (SI, SE, SO, test_mode, scan_clk, scan_rst)
 - Mux all the design clocks with scan_clk
 - Mux all the design resets with scan_rst
2. Open **cons.tcl** and Add the following commands: -
 - Clock Frequency with 1 MHz
 - Clock uncertainty with 0.025 ns for setup
 - Clock uncertainty with 0.01 ns for hold
 - Input delays on all input ports except (CLK & RST) with 30% clock period
 - Output delays on all output ports with 30% clock period
 - Add Buffer driving cell for all input ports
 - Add load of 0.1 pf on all output ports
 - Add dont_touch attribute on CLK & RST ports
 - Set operation condition using slow and fast libraries
3. Open **dft_script.tcl** and Add the following commands: -
 - 1) Read UART TX Verilog files
 - 2) Configuring scan chain style using **set_scan_configuration** command
 - 3) Test-Ready Compile using **compile -scan** command
 - 4) Defining the DFT Signals (SI, SE, SO, test_mode, scan_clk, scan_rst) using **set_dft_signal** command
 - 5) Create Test Protocol using **create_test_protocol** command
 - 6) Perform Pre-DFT Design Rule Checking using **dft_drc -verbose** command
 - 7) Preview DFT using **preview_dft -show scan_summary** command
 - 8) Insert DFT logic using **insert_dft** command

- 9) Optimize Post DFT Insertion using `compile -scan -incremental` command
 - 10) Perform Post-DFT Design Rule Checking using `dft_drc -verbose -coverage_estimate` command
 - 11) Write Verilog netlist using (`write_file -format verilog -hierarchy`) command
 - 12) Generate Power report using (`report_power -hierarchy`) command
 - 13) Generate Area report using (`report_area -hierarchy`) command
 - 14) Generate Setup timing analysis report using (`report_timing -max_paths 100 -delay_type max`) command
 - 15) Generate Hold timing analysis report using (`report_timing -max_paths 100 -delay_type min`) command
 - 16) Generate Clocks report using (`report_clock -attributes`) command
 - 17) Generate Constraints report using (`report_constraint -all_violators`) command
4. Run `dft_script.tcl` and Check the dft log (`dft.log`) is free of (`Errors, Latches, Comb Loops`) and modify your RTL files to fix any issues.

Required: -

A) You have to deliver the following files: -

- 0- dft script >> `dft_script.tcl`
- 1- dft log >> `dft.log`
- 2- Post DFT Inserted Gate Level Netlist >> `UART_TX.v`
- 3- Area report >> `Area.rpt`
- 4- Power report >> `power.rpt`
- 5- Hold analysis report >> `hold.rpt`
- 6- Setup analysis report >> `setup.rpt`
- 7- Clocks report >> `clocks.rpt`
- 8- Constraints report >> `constraints.rpt`
- 9- Ports report >> `ports.rpt`
- 10- Coverage Report >> `dft_drc_post_dft.rpt`